

Notes

- **Assignments**

- **3.2**

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- **Tutorial problems**

- **Basic Problems with Answers 3.11**

- **Basic Problems 3.30, 3.37**

- **Advanced Problems 3.49**

CT Fourier Series Pairs

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{j2\pi kt/T}$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

- Frequency components are harmonically related
- There exists a convergence issue on the synthesis equation
- Analysis equation is a projection on the bases of periodic signal space

Periodicity Properties of DT Complex Exponentials

- For DT complex exponentials, signals are periodic only when $\omega_0 N = k \cdot 2\pi$, $k = 0, \pm 1, \pm 2, \dots$

$$e^{j\omega_0 n} = e^{j\omega_0 (n+N)} \rightarrow e^{j\omega_0 N} = 1 \rightarrow \omega_0 N = k \cdot 2\pi$$

- For DT complex exponentials, signals with frequencies ω_0 and $\omega_0 + k \cdot 2\pi$ are identical.
$$e^{j(\omega_0 + k \cdot 2\pi)n} = e^{j\omega_0 n} \cdot e^{jk \cdot 2\pi n} = e^{j\omega_0 n}$$
 - We need only consider a frequency interval of length 2π , and in most cases, we use the interval: $0 \leq \omega_0 < 2\pi$, or $-\pi \leq \omega_0 < \pi$

- $e^{j\omega_0 n}$ does **not** have a continually increasing rate of oscillation as ω_0 is increased in magnitude.

low-frequency (slowly varying): ω_0 near $0, 2\pi, \dots$, or $2k \cdot \pi$

high-frequency (rapid variation): ω_0 near $\pm \pi, \dots$, or $(2k+1) \cdot \pi$

$$e^{j(2k+1)\pi n} = e^{j\pi n} = (e^{j\pi})^n = (-1)^n$$

DT Fourier Series Representation

Arbitrary periodic DT signal with period N can be written as

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$\sum_{k=\langle N \rangle}$ = Sum over *any* N consecutive values of k

— This is a *finite* series

$\{a_k\}$ - Fourier (series) coefficients

Frequency components: $\frac{2k\pi}{N}$ $k = 0, 1, 2, \dots, N-1$ or $1, 2, \dots, N$

Why?

Existence

Any DT periodic signal has a Fourier series representation

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n} \quad \forall n$$

$\Downarrow$

$n=0$

$$x[0] = \sum_{k=\langle N \rangle} a_k$$

$n=1$

$$x[1] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0}$$

$n=2$

$$x[2] = \sum_{k=\langle N \rangle} a_k e^{j2k\omega_0}$$

$\vdots$

$\vdots$

$\vdots$

$n=N-1$

$$x[N-1] = \sum_{k=\langle N \rangle} a_k e^{j(N-1)k\omega_0}$$

N equations for N unknowns, $a_0, a_1, \dots, a_{N-1}$

How to calculate a_k

- Define inner product of signals with period N as

$$\langle e^{jk\omega_0 n} \cdot e^{jm\omega_0 n} \rangle = \frac{1}{N} \sum_{n=0}^{N-1} e^{jk\omega_0 n} e^{-jm\omega_0 n}$$

- We have

$$\langle e^{jk\omega_0 n} \cdot e^{jm\omega_0 n} \rangle = 1 \quad (k = m + Nk')$$

$$\langle e^{jk\omega_0 n} \cdot e^{jm\omega_0 n} \rangle = 0 \quad (\textit{Otherwise})$$

- $\{e^{jk\omega_0 n} | k = \langle N \rangle\}$ is the orthonormal basis of DT periodic signal space with period N

- Because

$$\begin{aligned}
 \langle x[n] \cdot e^{jk\omega_0 n} \rangle &= \langle \sum_{m=0}^{N-1} a_m e^{jm\omega_0 n} \cdot e^{jk\omega_0 n} \rangle \\
 &= \sum_{m=0}^{N-1} a_m \langle e^{jm\omega_0 n} \cdot e^{jk\omega_0 n} \rangle = a_k
 \end{aligned}$$

- Hence,

$$a_k = \langle x[n] \cdot e^{jk\omega_0 n} \rangle = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n}$$

DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_o n} \quad (\text{Synthesis equation})$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_o n} \quad (\text{Analysis equation})$$

Different
from CT
Fourier
series

Cont.

- a_k can be defined for all integers k , and we have $a_{k+N} = a_k$

$$x[n] = a_0 e^{\frac{j0 \times 2\pi}{N}n} + a_1 e^{\frac{j1 \times 2\pi}{N}n} + \dots + a_{N-1} e^{\frac{j(N-1) \times 2\pi}{N}n}$$

$$x[n] = a_1 e^{\frac{j1 \times 2\pi}{N}n} + \dots + a_{N-1} e^{\frac{j(N-1) \times 2\pi}{N}n} + a_N e^{\frac{jN \times 2\pi}{N}n}$$

- a_k is periodic w.r.t. k
- CT is different

Example #1: Sum of a pair of sinusoids

$$x[n] = \cos(\pi n / 8) + \cos(\pi n / 4 + \pi / 4)$$

— periodic with period $N = ?$

$$x[n] = \frac{1}{2} [e^{j\omega_0 n} + e^{-j\omega_0 n}] + \frac{1}{2} [e^{j\pi/4} e^{j2\omega_0 n} + e^{-j\pi/4} e^{-j2\omega_0 n}]$$

$\Downarrow$

$$a_0 = 0$$

$$a_{15} = a_{-1+16} = a_{-1} = 1/2$$

$$a_1 = 1/2$$

$$a_{-1} = 1/2$$

$$a_{66} = a_{2+4 \times 16} = a_2 = e^{j\pi/4}/2$$

$$a_2 = e^{j\pi/4}/2$$

$$a_{-2} = e^{-j\pi/4}/2$$

$$a_3 = 0$$

$$a_{-3} = 0$$

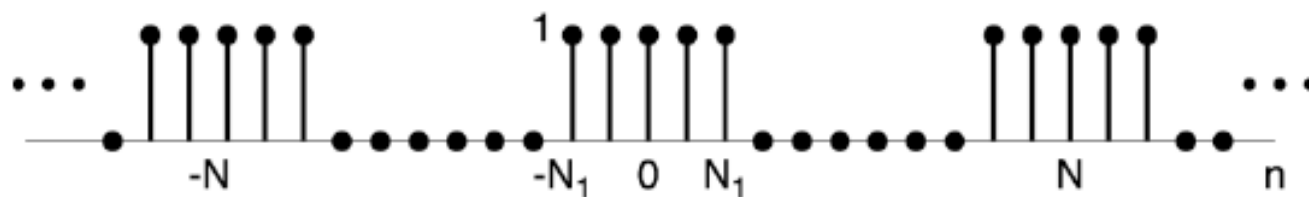
$\vdots$

$$\begin{aligned} \cos(x) &= \operatorname{Re}(e^{jx}) = \frac{1}{2} (e^{jx} + e^{-jx}) \\ \sin(x) &= \operatorname{Im}(e^{jx}) = \frac{1}{2} (e^{jx} - e^{-jx}) \end{aligned}$$

Example 3.12

DT Square wave

Period=?



$$a_0 = \frac{1}{N} \sum_{n=-N_1}^{N_1} x[n] = \frac{(2N_1 + 1)}{N} = a_N = a_{-N} = a_{6N} = \dots$$

For $k \neq$ multiple of N :

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=-N_1}^{N_1} e^{-jk\omega_0 n} \stackrel{n=m-N_1}{=} \frac{1}{N} \sum_{m=0}^{2N_1} e^{-jk\omega_0 (m-N_1)} \\ &= \frac{1}{N} e^{jk\omega_0 N_1} \sum_{m=0}^{2N_1} (e^{-jk\omega_0})^m = \frac{1}{N} e^{jk\omega_0 N_1} \frac{1 - e^{-jk\omega_0 (2N_1 + 1)}}{1 - e^{-jk\omega_0}} \\ &= \frac{1}{N} \frac{\sin[k(N_1 + 1/2)\omega_0]}{\sin(k\omega_0 / 2)} = \frac{1}{N} \frac{\sin[2\pi k(N_1 + 1/2) / N]}{\sin(\pi k / N)} \end{aligned}$$

DT Square wave (continued)

$$a_k = \frac{1}{N} \frac{\sin[2\pi k(N_1 + 1/2)/N]}{\sin(\pi k/N)}$$

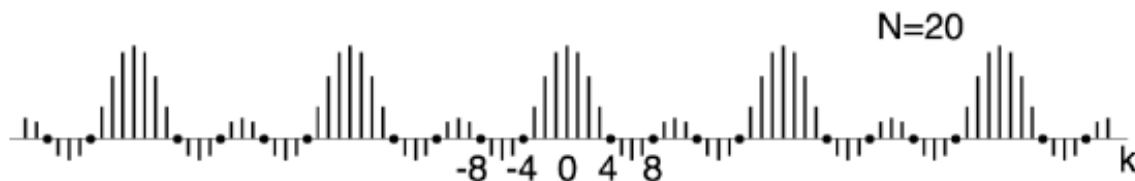
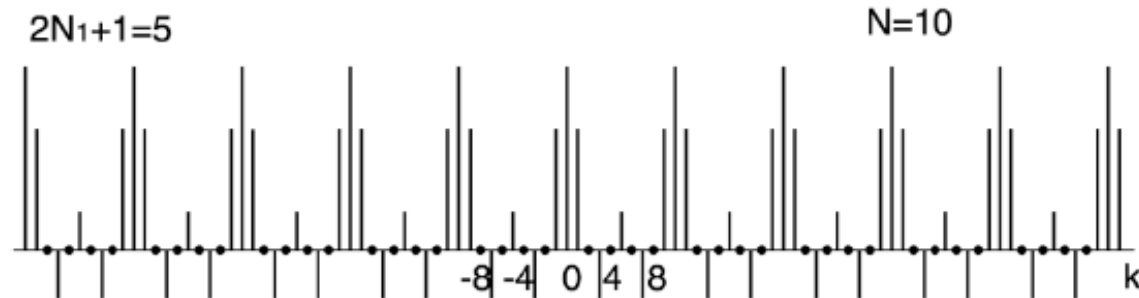
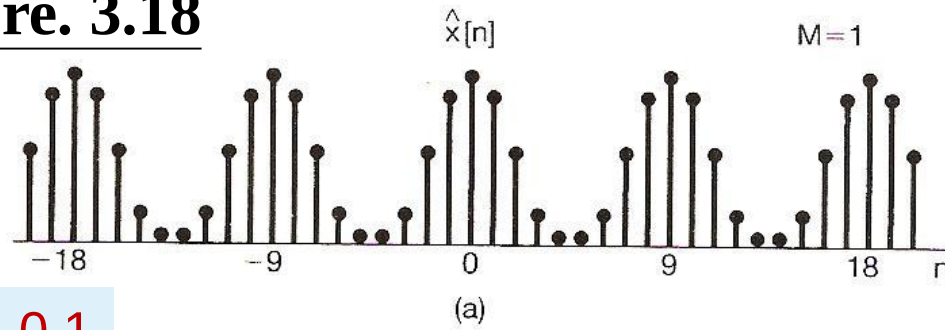


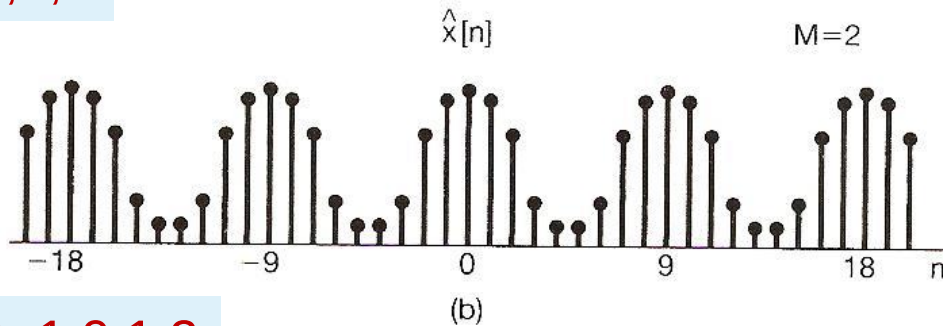
Figure. 3.18

$$N=9, 2N_1+1=5$$

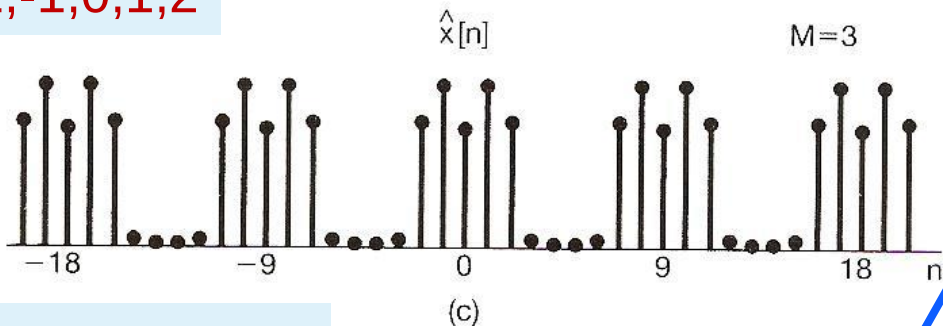
$$k=-1,0,1$$



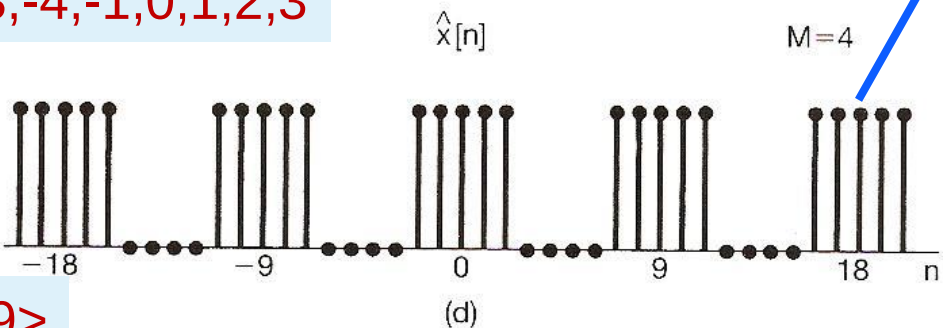
$$k=-2,-1,0,1,2$$



$$k=-3,-4,-1,0,1,2,3$$



$$k=<9$$



- 1) The same as original DT square wave
- 2) **No** Gibbs phenomenon, and **no** discontinuity

Figure 3.18 Partial sums of eqs. (3.106) and (3.107) for the periodic square wave of Figure 3.16 with $N = 9$ and $2N_1 + 1 = 5$: (a) $M = 1$; (b) $M = 2$; (c) $M = 3$; (d) $M = 4$.

DT Fourier Series - Properties

- Strong similarities between the properties of DT and CT Fourier series [Comparing Table 3.2 to Table 3.1.]
- For example
 - Fourier series of $x[n - n_0]$, $e^{jm\omega_0 n}x[n]$
 - Fourier series properties for real signal
 - Fourier series of the multiplication

Two Important Properties

- **Periodic convolution:**

Suppose x and y are two periodic signals with common period N , the periodic convolution between x and y is defined as

$$x[n] \circledast y[n] = \sum_{k=\langle N \rangle} x[k]y[n-k]$$

- Suppose $x[n] \rightarrow a_k$ and $y[n] \rightarrow b_k$, then

$$x[n] \circledast y[n] \rightarrow Na_k b_k \quad \text{and} \quad x[n]y[n] \rightarrow a_k \circledast b_k$$

- **Parseval's Relation**

$$\frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 = \sum_{k=\langle N \rangle} |a_k|^2$$

Frequency Behavior of LTI Systems

Complex Exponentials - The Only Eigenfunctions of *Any* LTI Systems

$\rightarrow x(t) = e^{st} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \int_{-\infty}^{+\infty} h(\tau) e^{s(t-\tau)} d\tau$

$y(t) = x(t) * h(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$

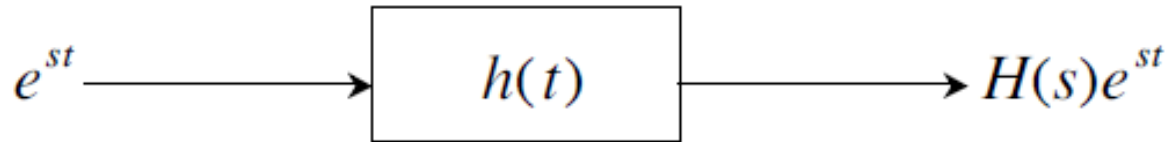
$$= \left[\int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau \right] e^{st}$$
$$= \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}}$$

$\rightarrow x[n] = z^n \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{m=-\infty}^{\infty} h[m] z^{n-m}$

$$= \left[\sum_{m=-\infty}^{\infty} h[m] z^{-m} \right] z^n$$
$$= \underbrace{H(z)}_{\text{eigenvalue}} \underbrace{z^n}_{\text{eigenfunction}}$$

Eigenfunction Property of Complex Exponentials

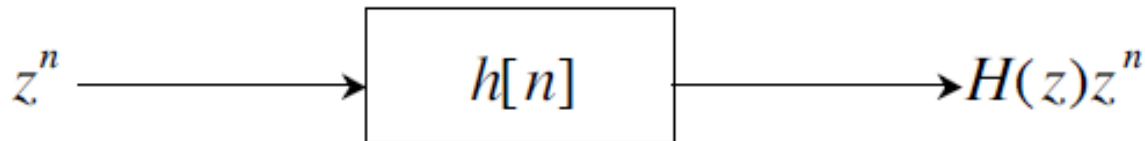
CT:



$$H(s) = \int_{-\infty}^{+\infty} h(t)e^{-st} dt$$

system function

DT:

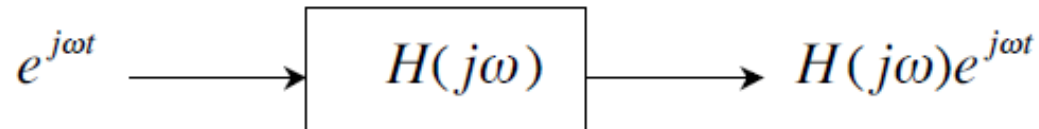


$$H(z) = \sum_{n=-\infty}^{+\infty} h[n]z^{-n}$$

system function

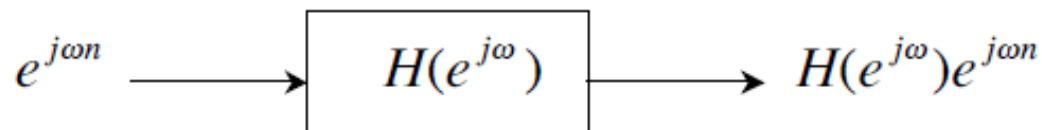
Frequency Response of an LTI System

$$(s = j\omega)$$



CT Frequency response:

$$H(j\omega) = \int_{-\infty}^{+\infty} h(t)e^{-j\omega t} dt$$



$$(z = e^{j\omega})$$

DT Frequency response:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n]e^{-j\omega n}$$

Periodic

Fourier Series and LTI Systems

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} H(jk\omega_0) a_k e^{jk\omega_0 t}$$

$$H(j\omega)$$

$$a_k \longrightarrow \underbrace{H(jk\omega_0) a_k}_{\text{"gain"}}$$

$$H(jk\omega_0) = |H(jk\omega_0)| e^{j\angle H(jk\omega_0)},$$

includes both amplitude & phase

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 n} \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} H(e^{jk\omega_0}) a_k e^{jk\omega_0 n}$$

$$H(e^{j\omega})$$

$$a_k \longrightarrow \underbrace{H(e^{jk\omega_0}) a_k}_{\text{"gain"}}$$

$$H(e^{jk\omega_0}) = |H(e^{jk\omega_0})| e^{j\angle H(e^{jk\omega_0})},$$

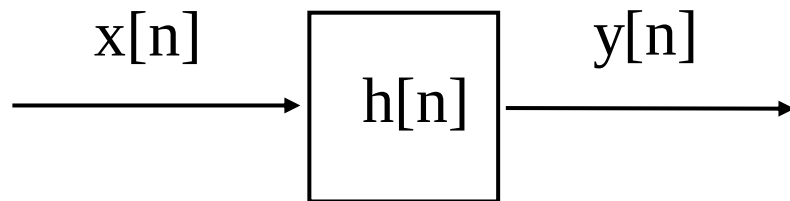
includes both amplitude & phase

The effect of the LTI system is to modify each a_k through multiplication by the value of the frequency response at the corresponding frequency.

Example 3.17

$$h[n] = \alpha^n u[n] \quad , \quad |\alpha| < 1$$

$$x[n] = \cos\left(\frac{2\pi n}{N}\right) = \frac{1}{2} e^{j(\frac{2\pi}{N})n} + \frac{1}{2} e^{-j(\frac{2\pi}{N})n}$$



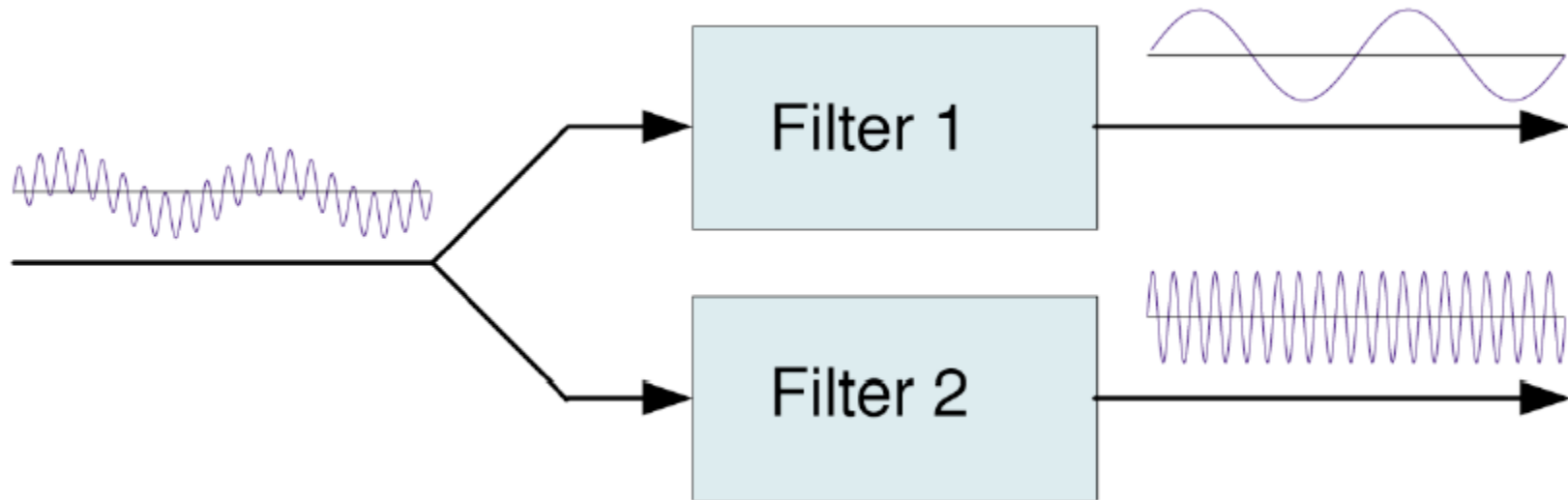
$$H(e^{j\omega}) = \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \frac{1}{1 - \alpha e^{-j\omega}}$$

$$\begin{aligned} y[n] &= \frac{1}{2} H\left(e^{j\frac{2\pi}{N}}\right) e^{j(\frac{2\pi}{N})n} + \frac{1}{2} H\left(e^{-j\frac{2\pi}{N}}\right) e^{-j(\frac{2\pi}{N})n} \\ &= r \cos\left(\frac{2\pi n}{N} + \theta\right) \end{aligned}$$

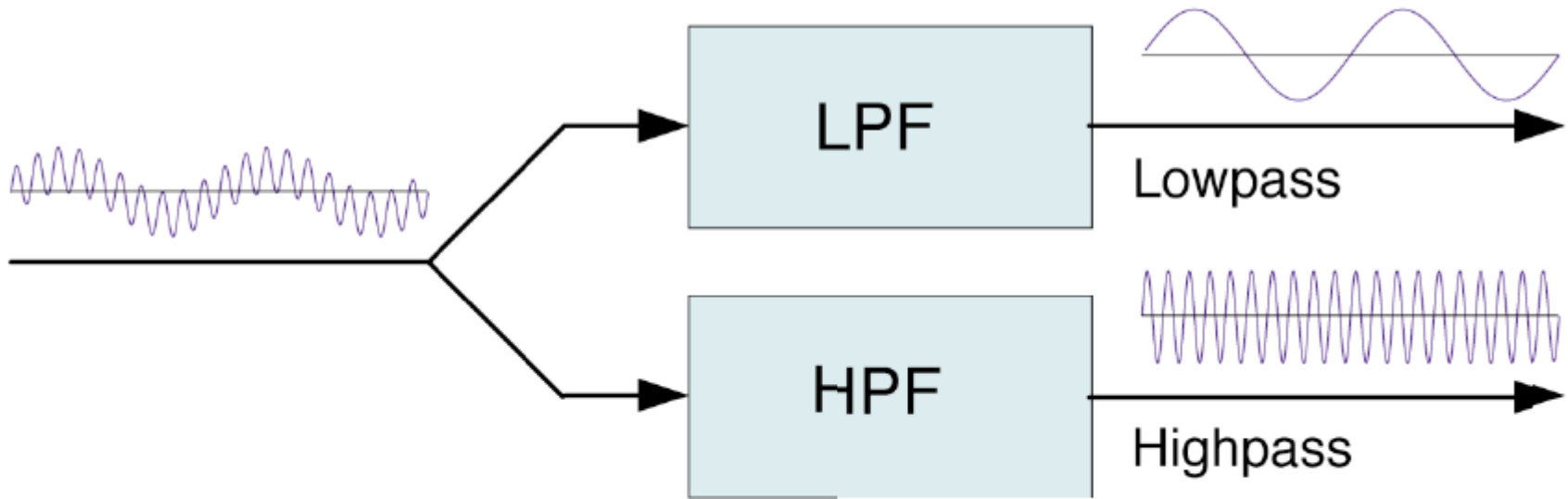
$$\text{where } re^{j\theta} = \frac{1}{1 - \alpha e^{-j\frac{2\pi}{N}}}$$

Signal Processing with Filters

- Separation of narrowband signals
 - Filter design — a main task in constructing communication systems.



Example: Using LPF and HPF to Separate Signals



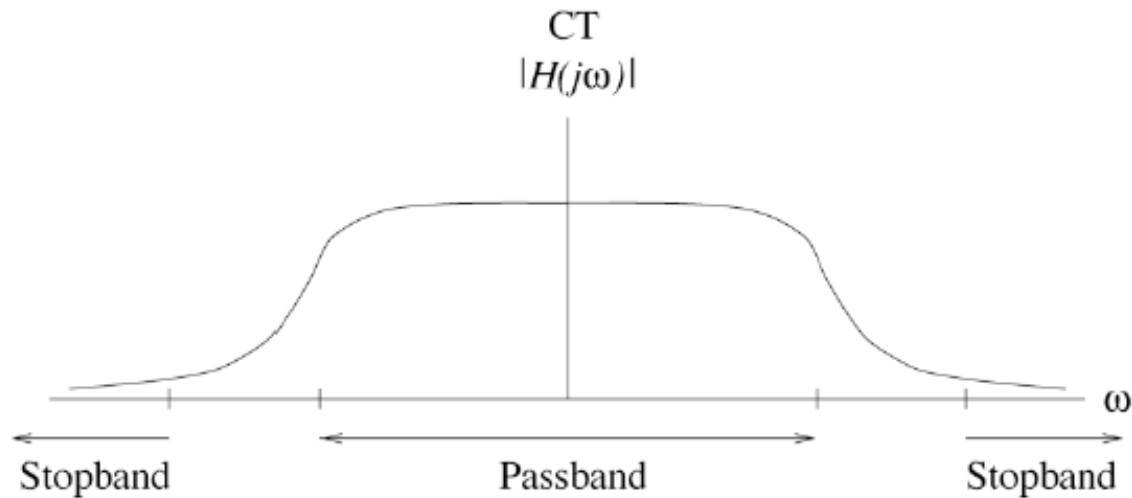
Filter design becomes more challenging if the two signals are close in frequency range.

Extraction of Narrowband Signals from Wideband Noise – BPF

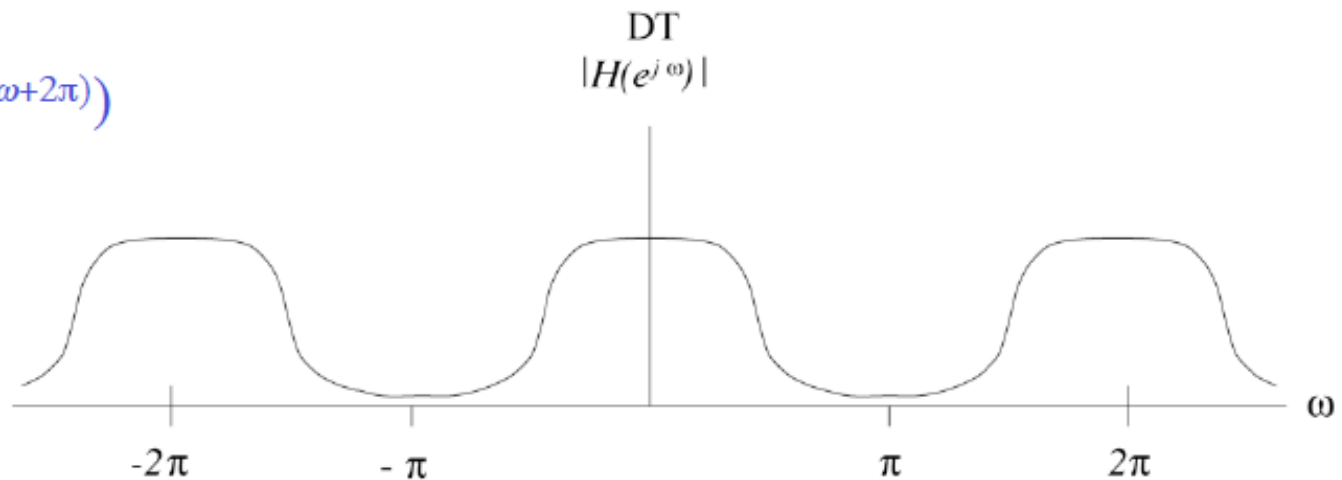


Lowpass Filter

Lowpass Filters:
Only show
amplitude here.

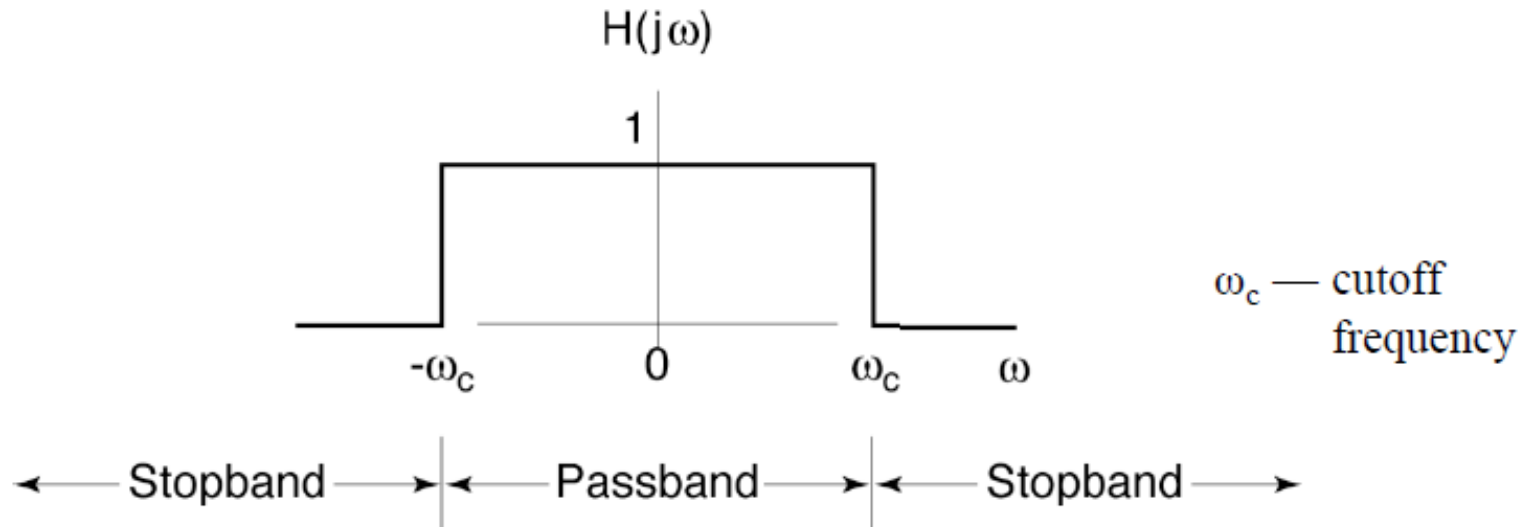


Note for DT:
 $H(e^{j\omega}) = H(e^{j(\omega+2\pi)})$

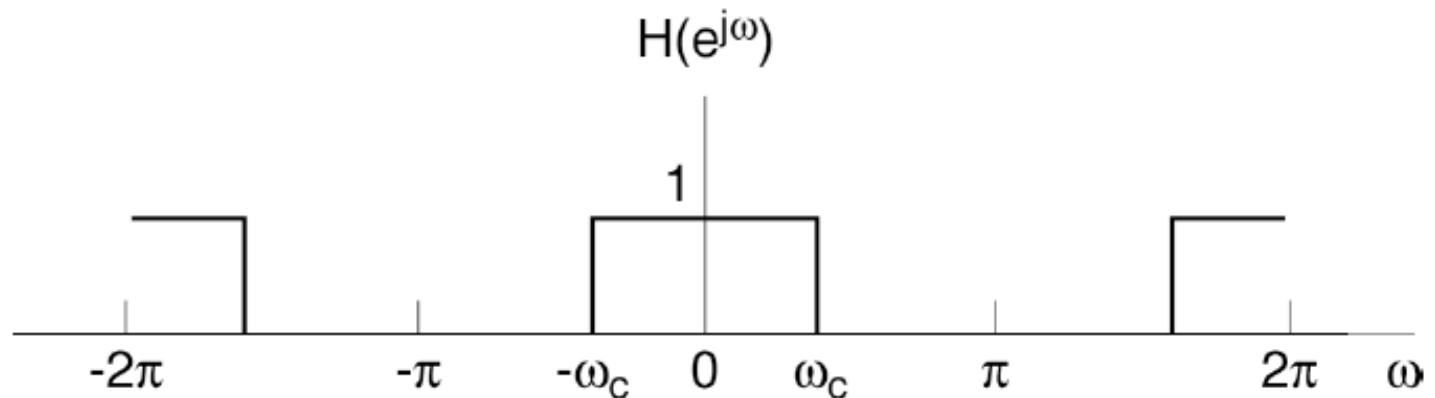


Ideal Lowpass Filter

CT



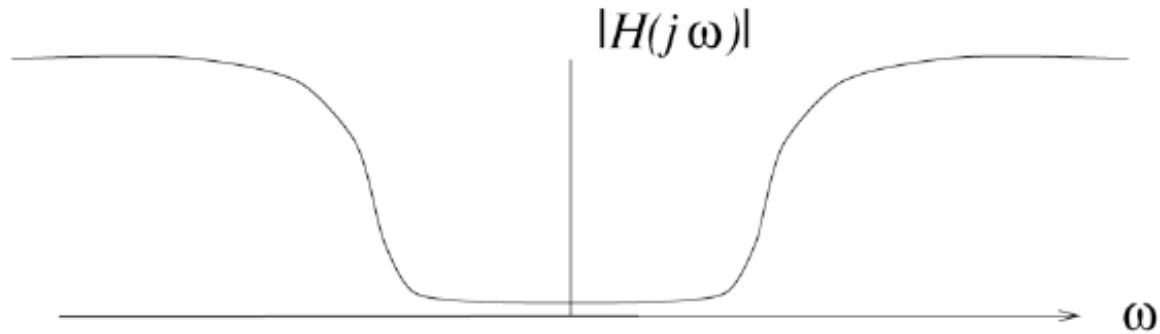
DT



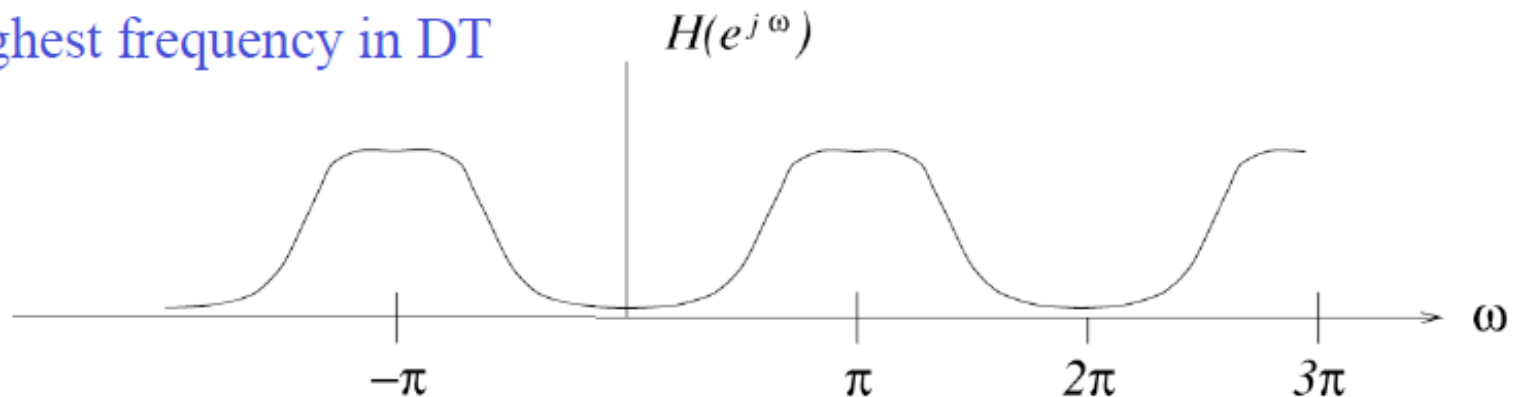
Note: $|H| = 1$ and $\angle H = 0$ for the ideal filters in the passbands, no need for the phase plot.

Highpass Filters

CT



DT



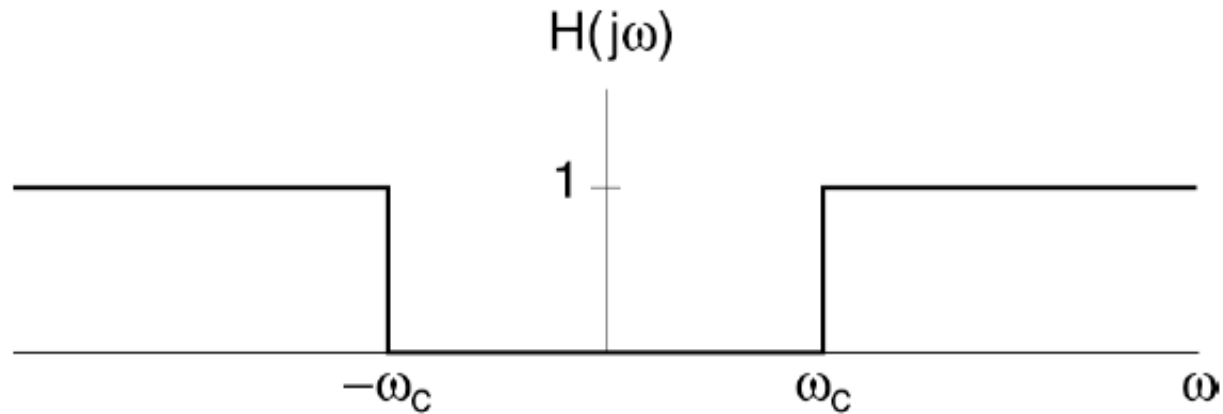
Remember:

$$(-1)^n = e^{j\pi n}$$

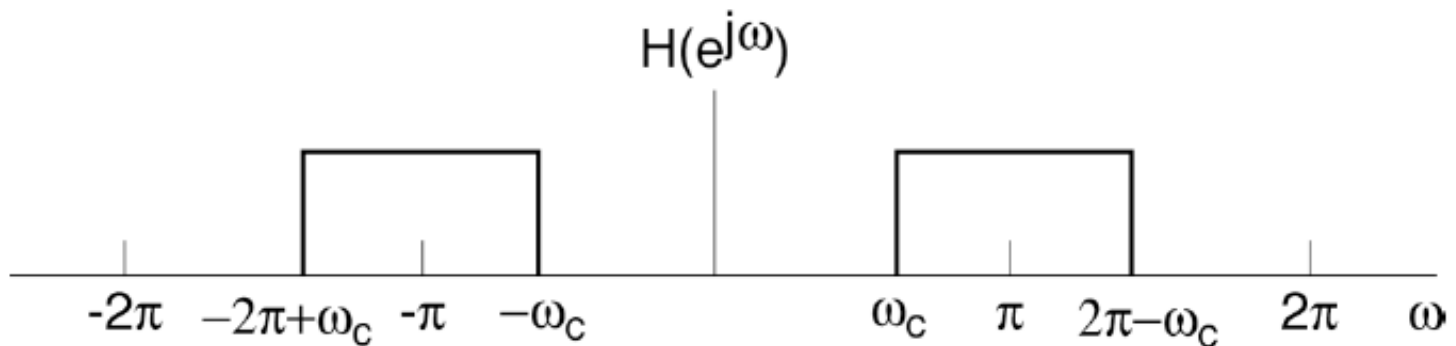
— π = highest frequency in DT

Ideal Highpass Filter

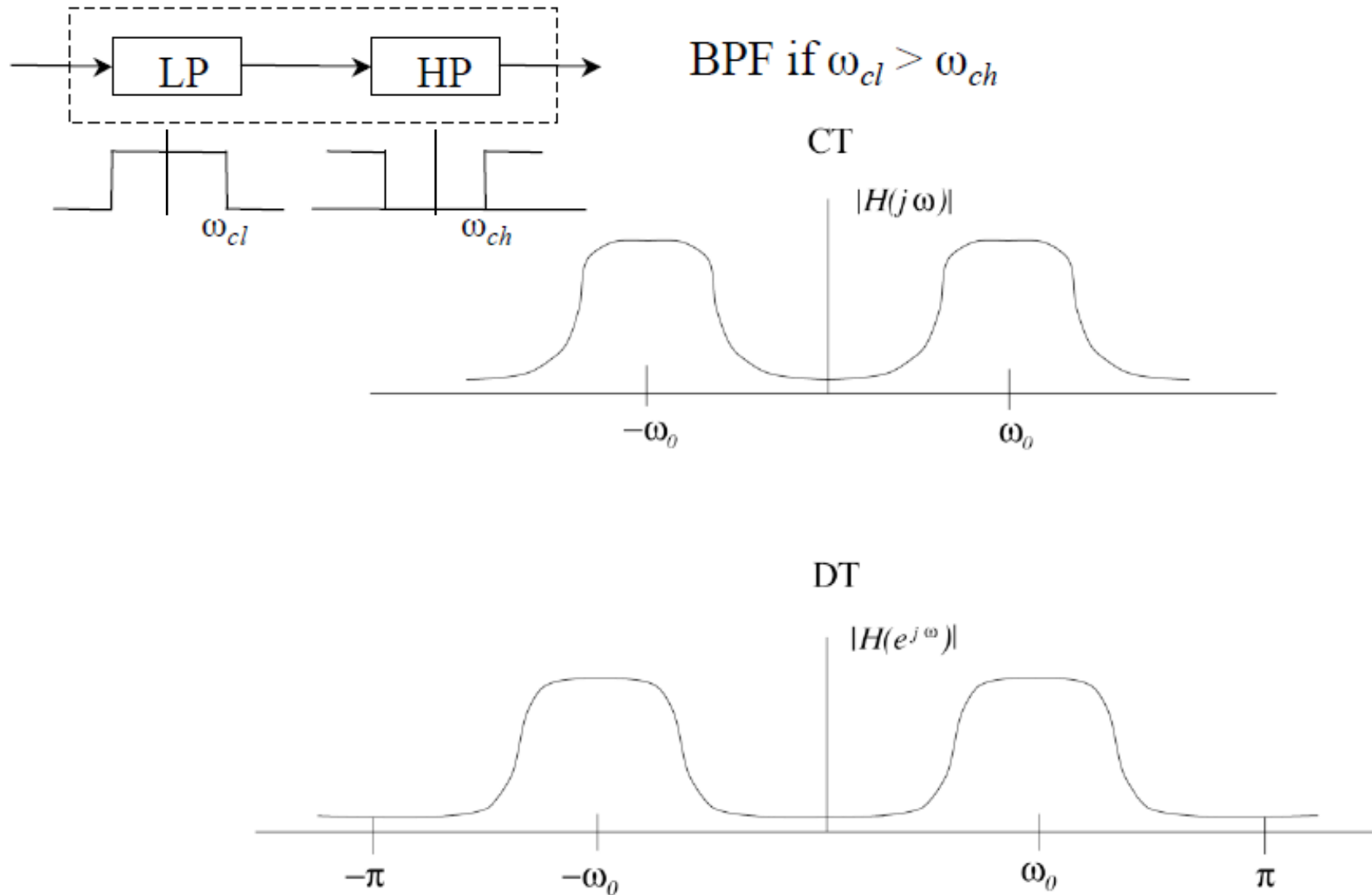
CT



DT

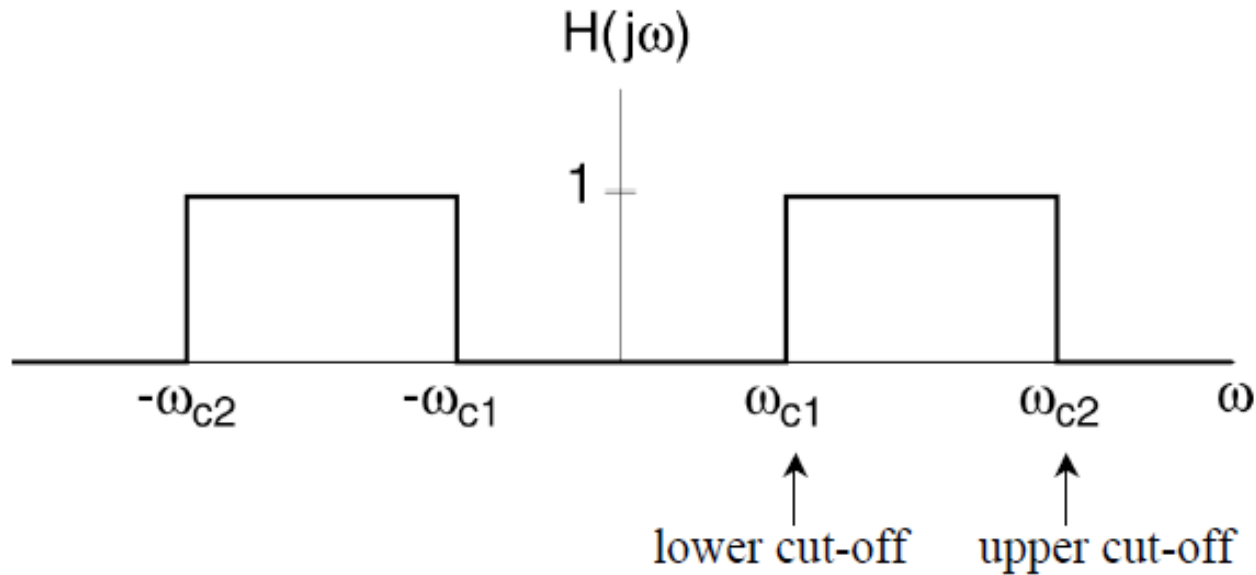


Bandpass Filters

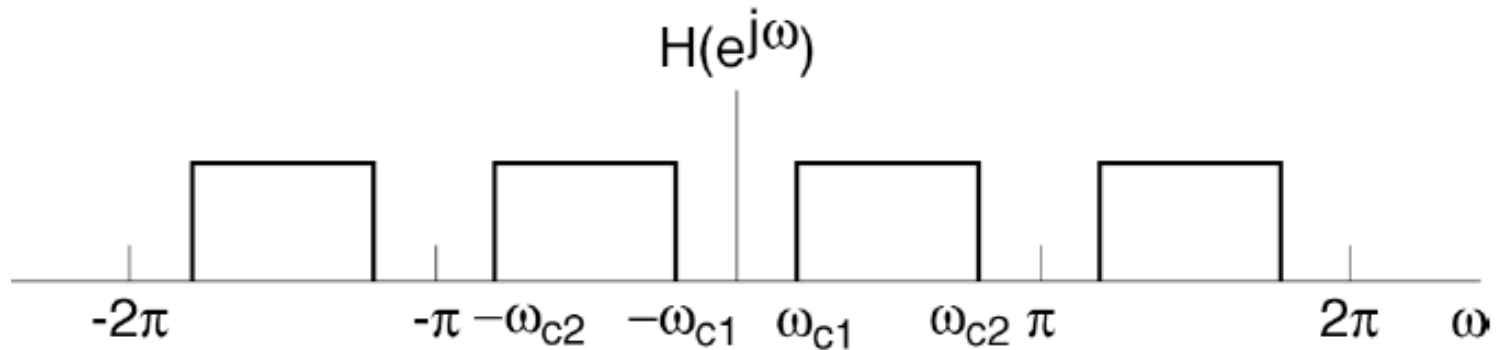


Ideal Bandpass Filter

CT



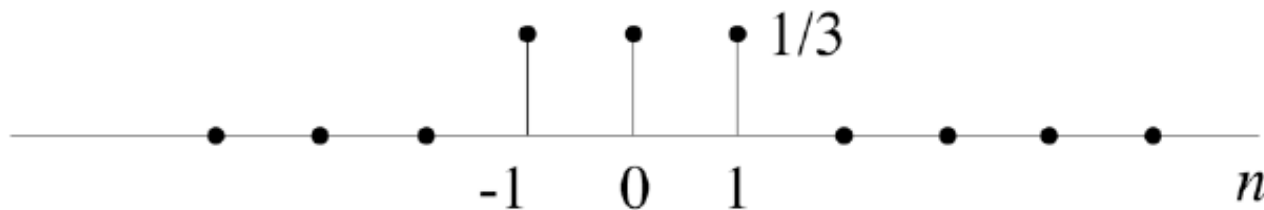
DT



Example: DT Averager/Smoothing

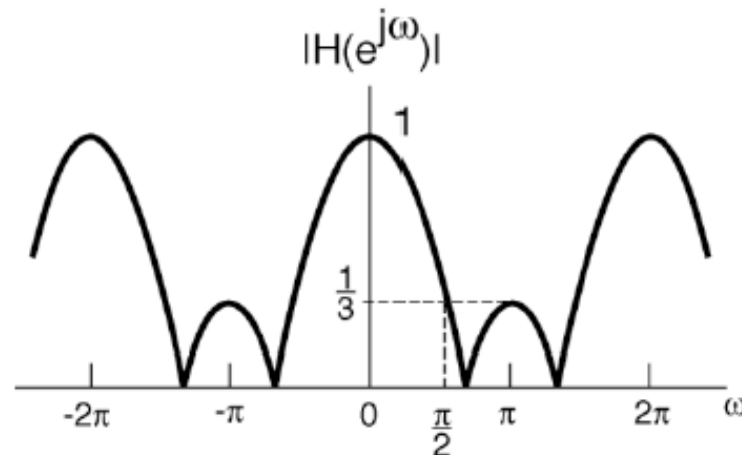
$$y[n] = \frac{1}{3} \{x[n-1] + x[n] + x[n+1]\}$$

$$h[n] = \frac{1}{3} \{\delta[n-1] + \delta[n] + \delta[n+1]\}$$



Frequency response:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \frac{1}{3}[e^{-j\omega} + 1 + e^{j\omega}] = \frac{1}{3} + \frac{2}{3}\cos\omega$$



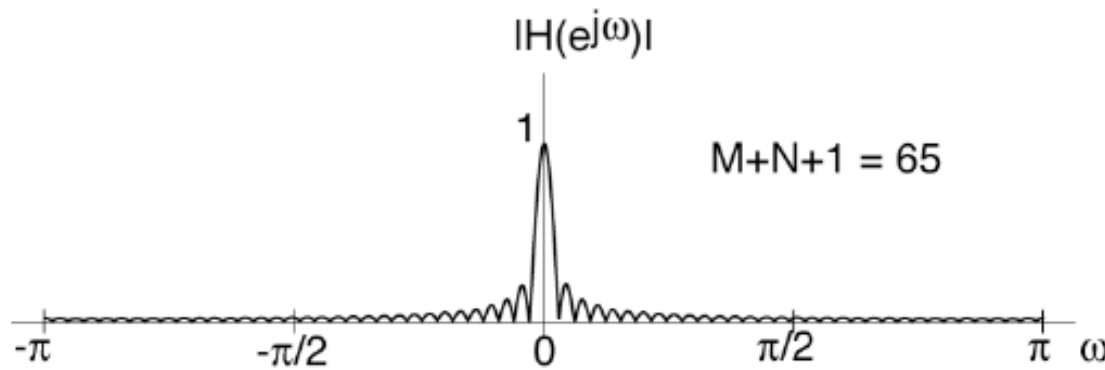
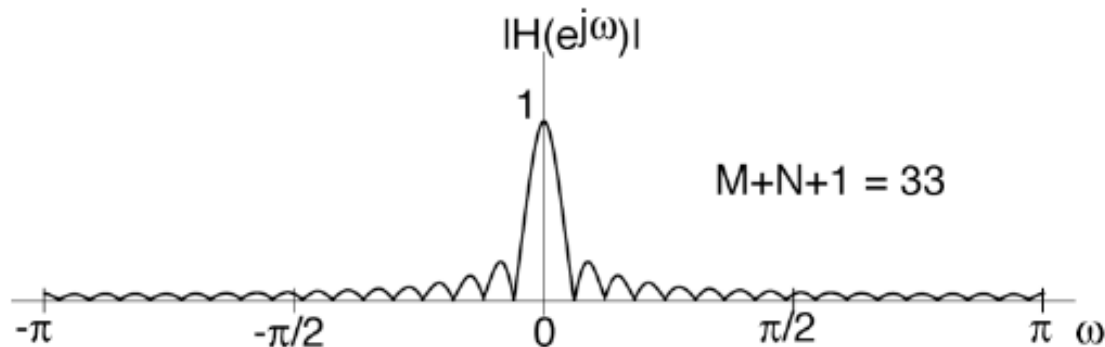
A LPF

Example: Nonrecursive DT (FIR) filters

$$y[n] = \frac{1}{N + M + 1} \sum_{k=-N}^M x[n - k] \longrightarrow h[n] = \frac{1}{N + M + 1} \sum_{k=-N}^M \delta[n - k]$$

Frequency response:

$$H(e^{j\omega}) = \frac{1}{N + M + 1} \sum_{k=-N}^M e^{-jk\omega} = \frac{1}{N + M + 1} e^{j\omega(N-M)/2} \frac{\sin[\omega(M + N + 1) / 2]}{\sin(\omega / 2)}$$



Rolls off at lower ω as $M+N+1$ increases

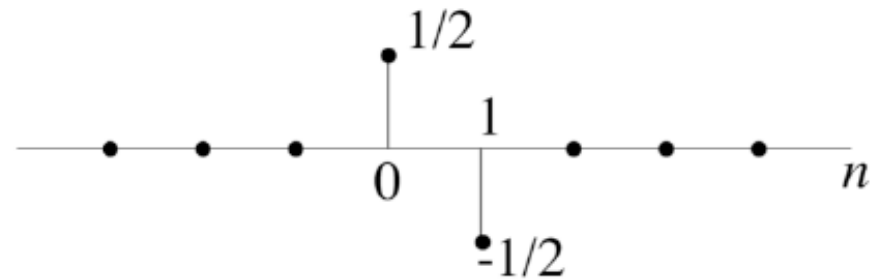
Example

Simple DT “Edge” Detector

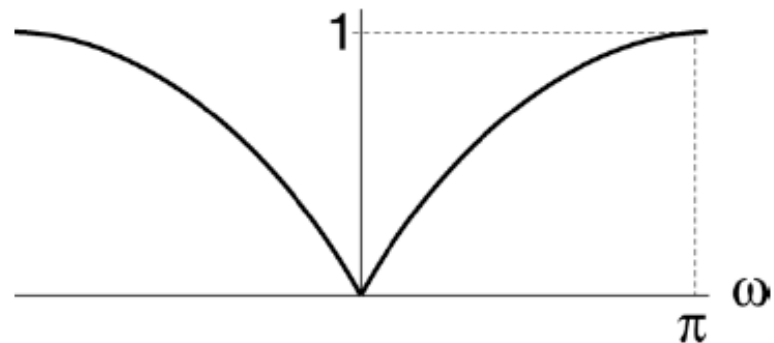
— DT 2-points “differentiator”

$$y[n] = \frac{1}{2}[x[n] - x[n-1]]$$

$$h[n] = \frac{1}{2}[\delta[n] - \delta[n-1]]$$



$|H(e^{j\omega})|$



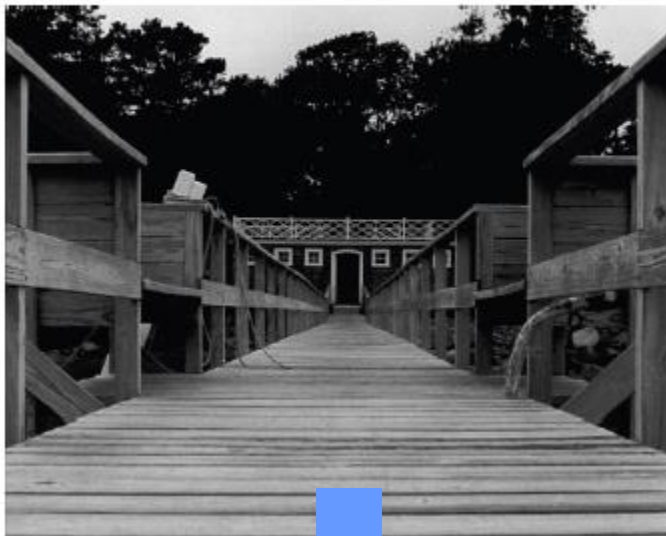
Amplifies high-frequency components

Frequency response:

$$H(e^{j\omega}) = \frac{1}{2}(1 - e^{-j\omega}) = je^{j\omega/2} \sin(\omega / 2)$$

$$|H(e^{j\omega})| = |\sin(\omega / 2)|$$

Edge enhancement using DT differentiator



Summary

- **DT Fourier Series pair**
 - Understand the difference between CT and DT
- **Frequency response**
 - How to determine frequency response?
- **Filtering**